Quantum Communication in Optical Networks: Enabling the Next Generation of Secure Connectivity

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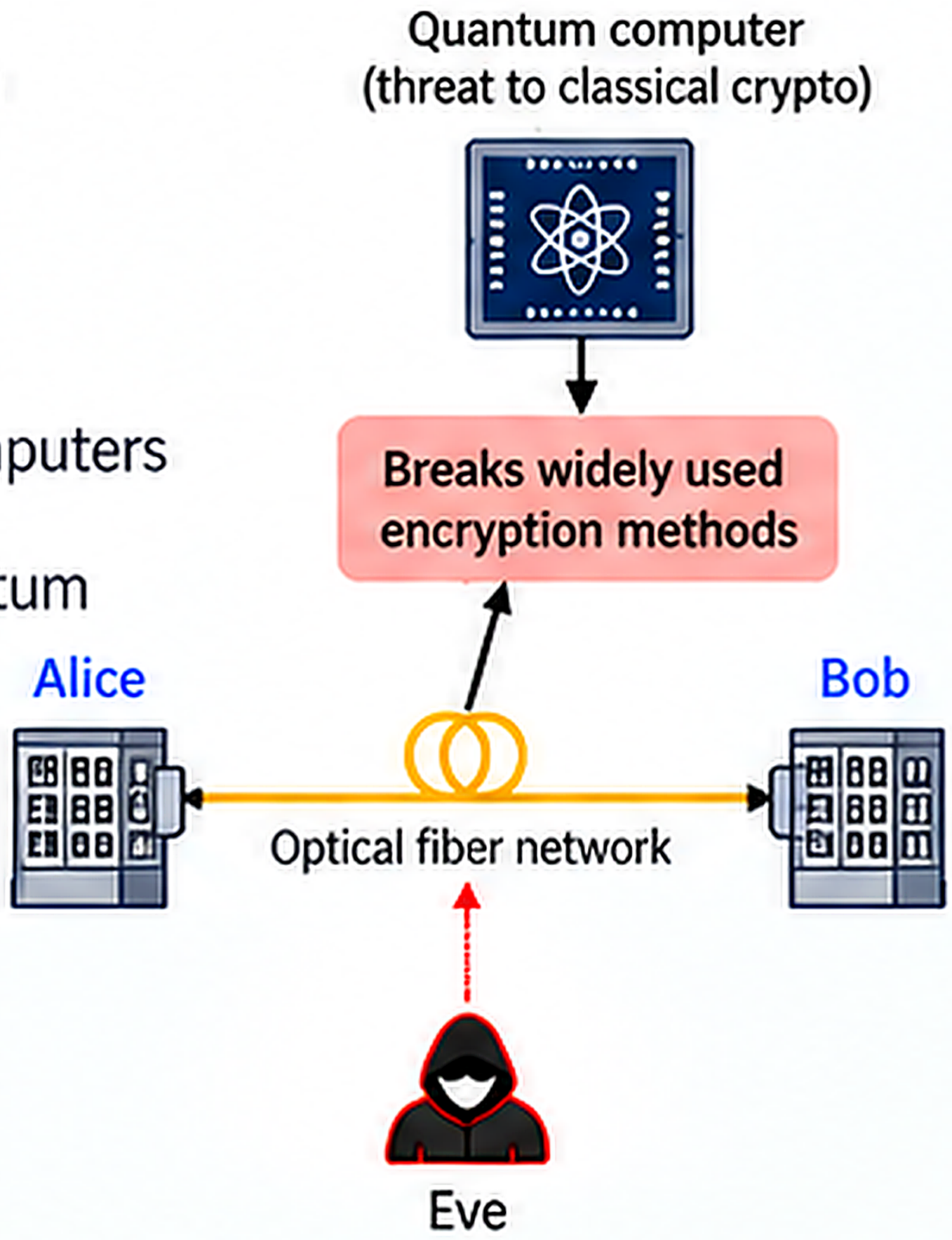
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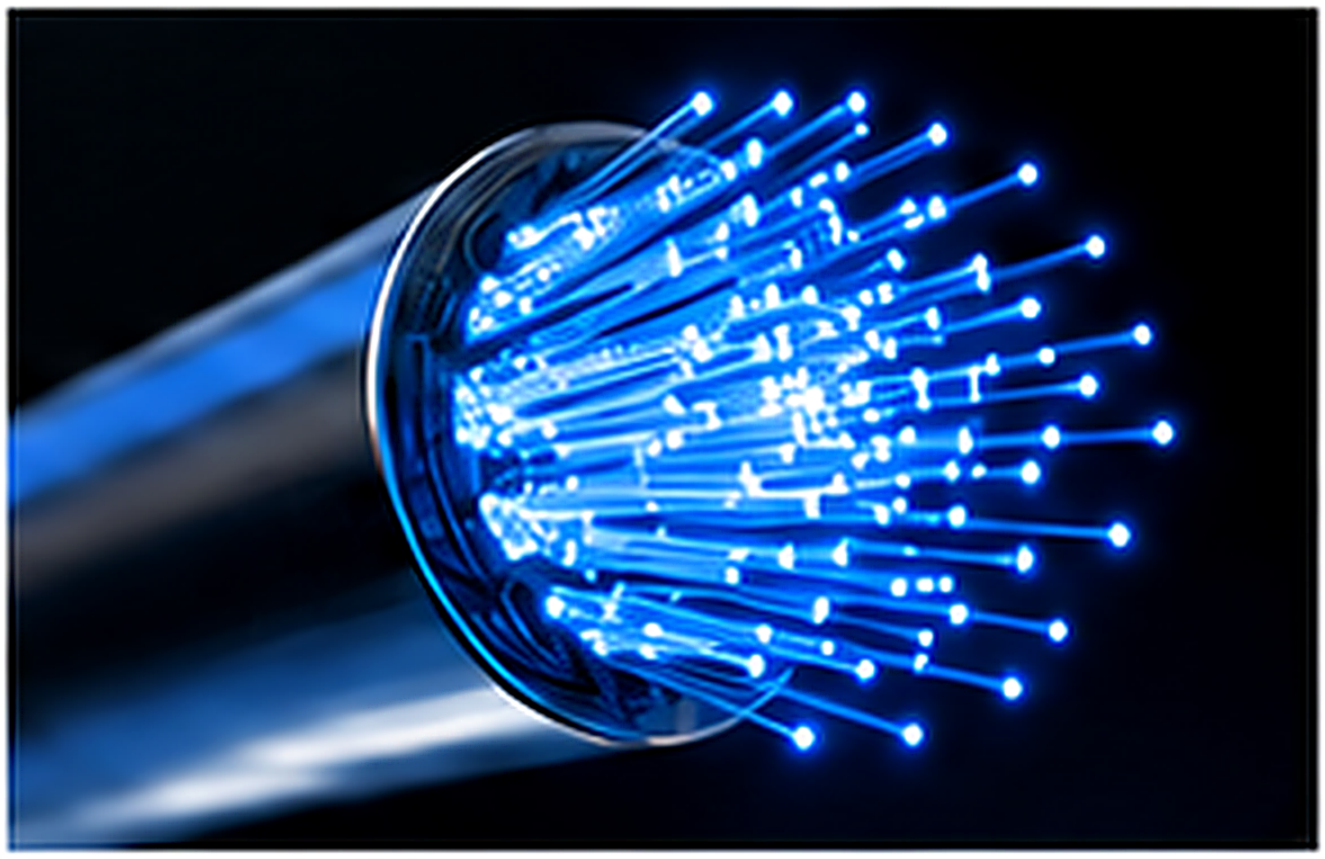
1. Why Quantum Key Distribution?

- Secure communication is essential
- Classical encryption may be vulnerable to future quantum computers
- QKD uses laws of quantum physics to establish secret keys
- Provides information-theoretic security

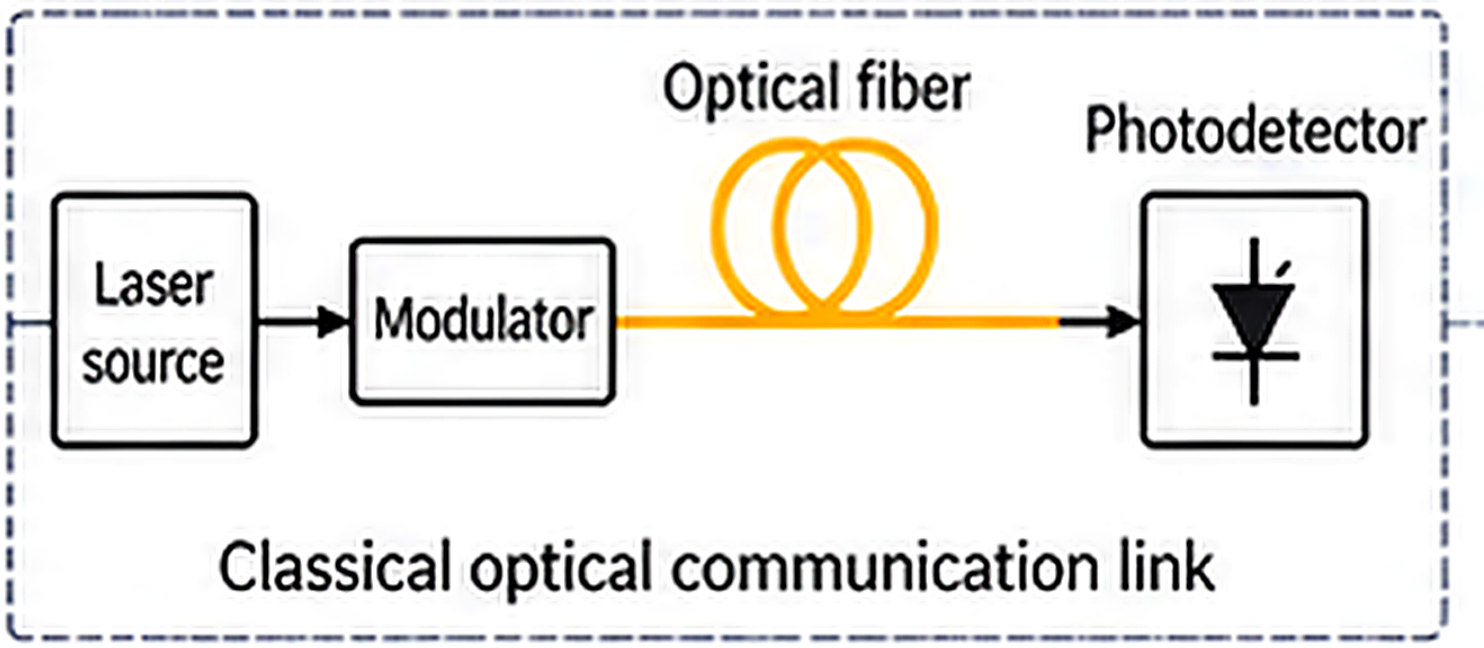


2. Optical Communication Networks

- Information carried by light in optical fibers
- Data encoded in intensity/phase/polarization
- Key components: laser, modulators, fiber, splitters, detectors

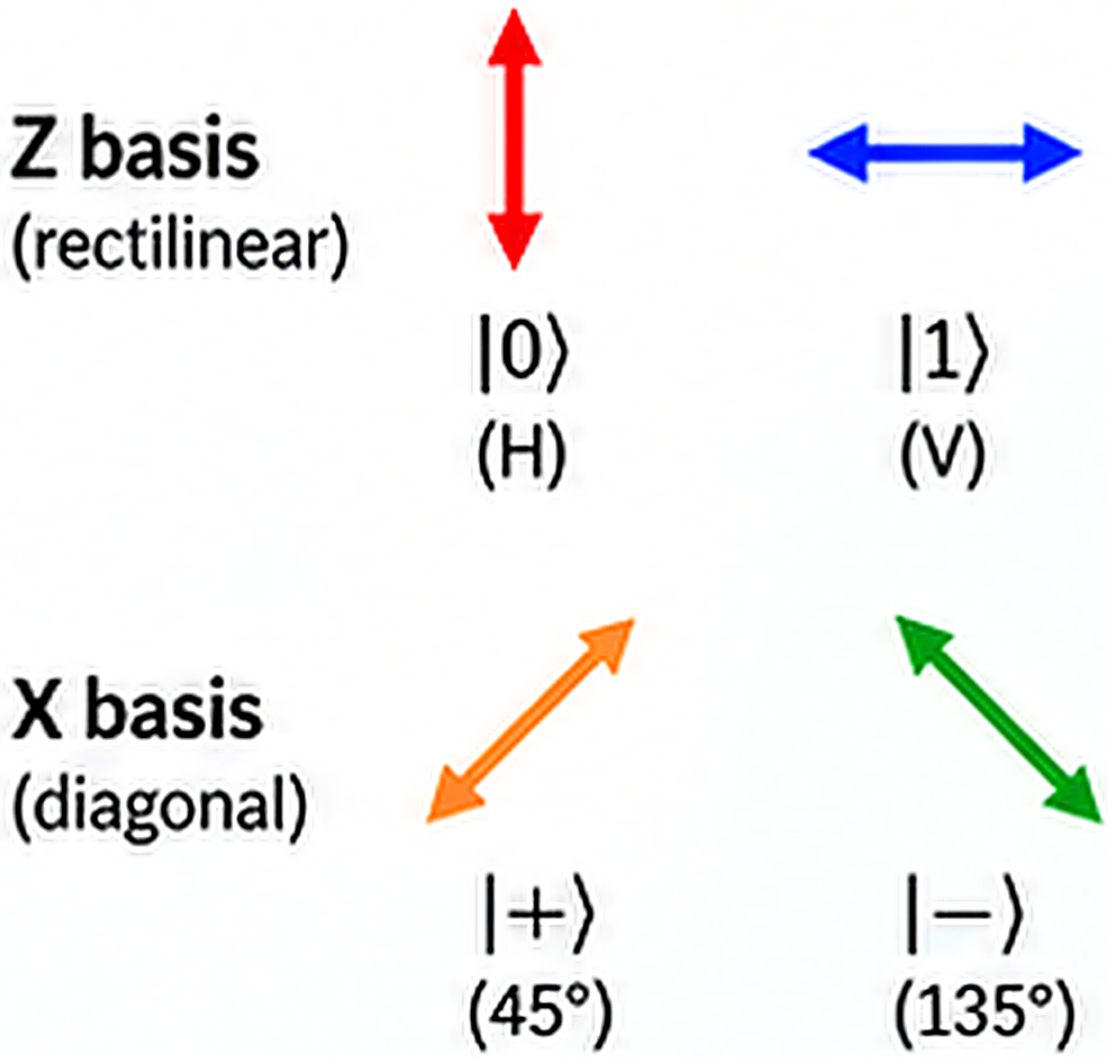


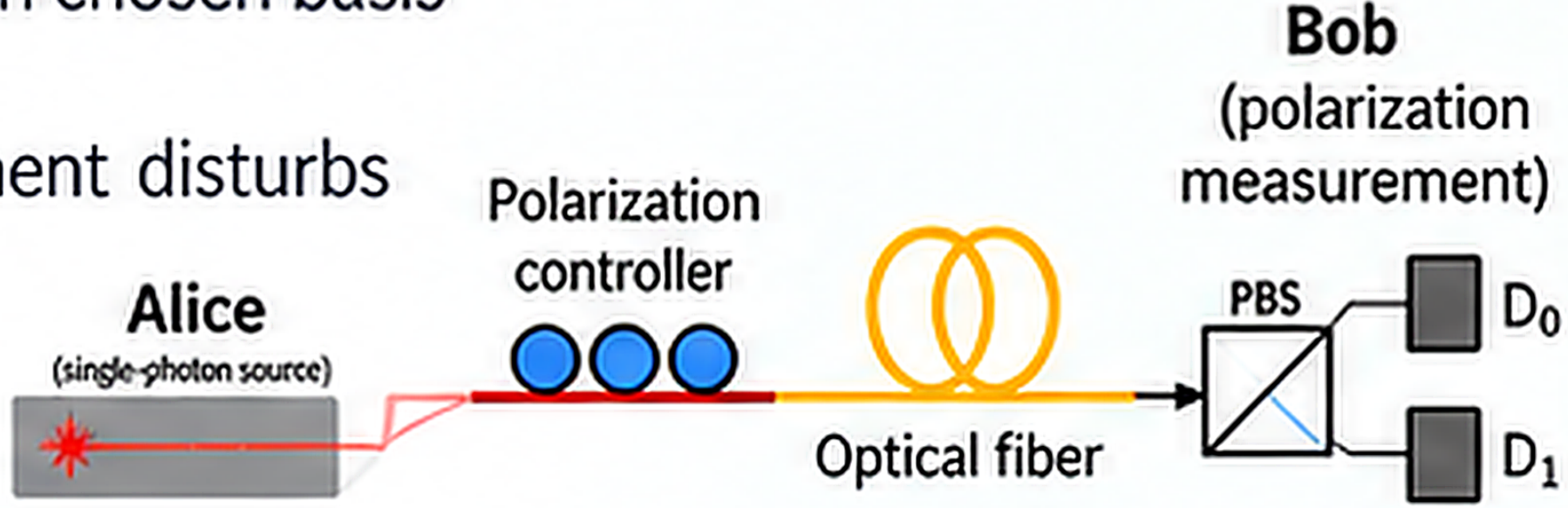
- High bandwidth and low loss
- QKD can share the same fiber infrastructure



3. Quantum States of Light

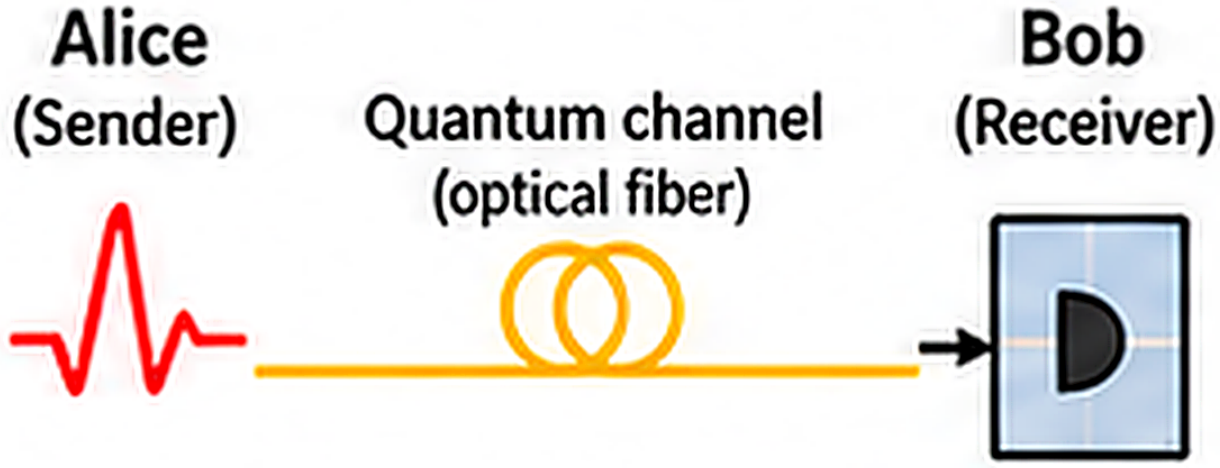
- Single photons or weak laser pulses
- Polarization states encode qubits
- Two conjugate bases (e.g., Z and X bases)
- Measurement outcome depends on chosen basis
- Measurement disturbs the state





4. BB84 Quantum Key Distribution

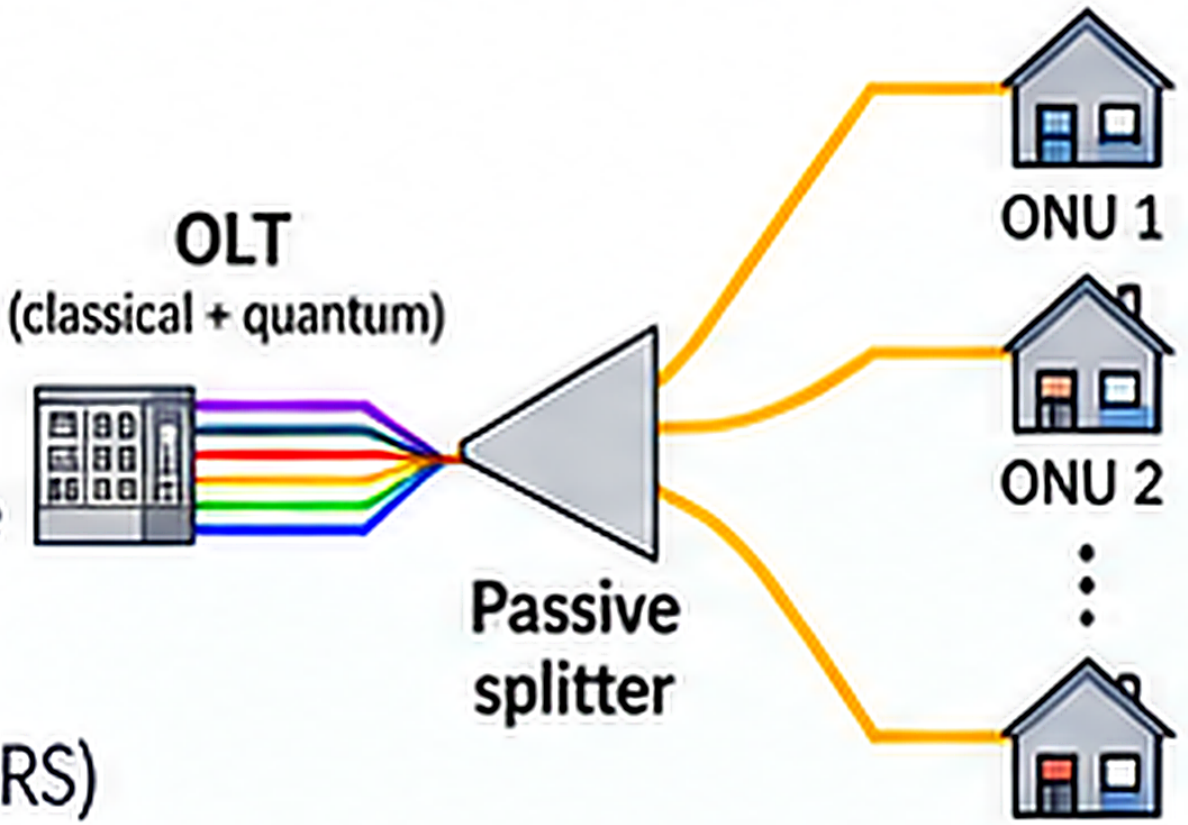
- Alice sends random bits in random bases
- Bob measures in random bases
- Publicly compare bases (keep matching ones)
- Test a subset for errors (eavesdropping check)
- Classical post-processing (error correction + privacy amplification)
- Result: shared secret key

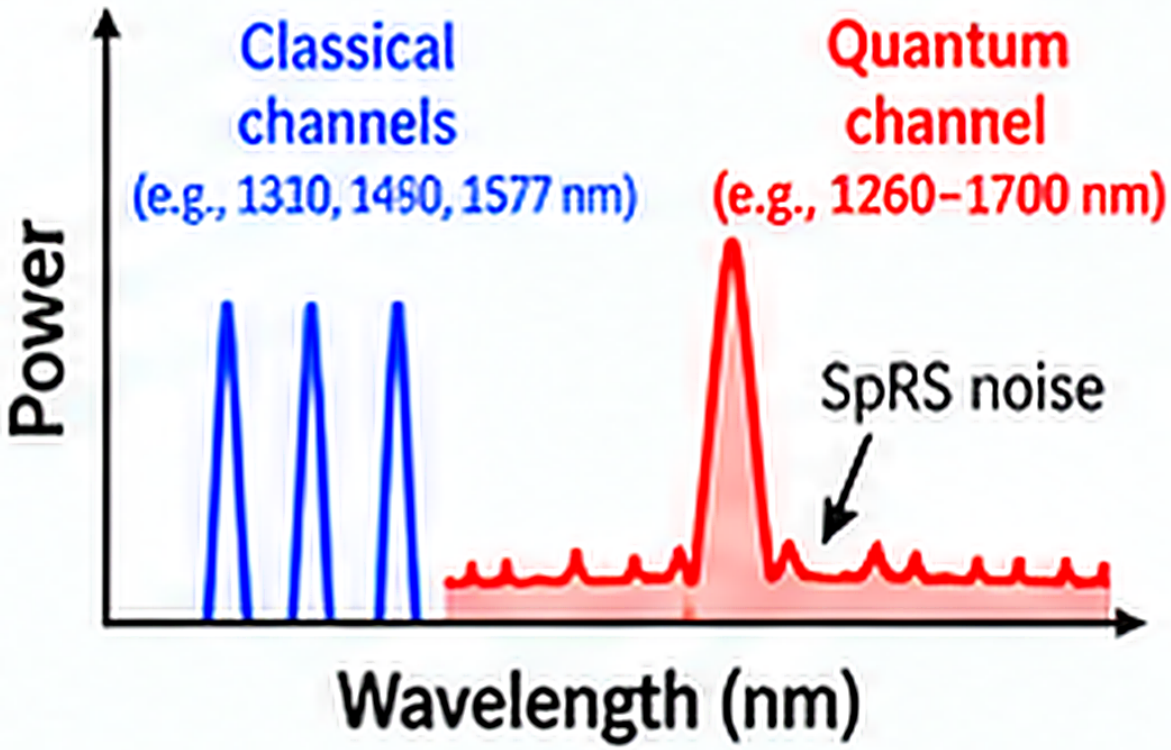


Step	Alice	Bob	Action / Result
1. Prepare	Random bit (0 or 1)	–	Choose random basis (Z or X)
2. Send	Send photon in chosen state	–	Transmit over optical fiber
3. Measure	–	Random basis (Z or X)	Measure photon
4. Sift	Announce bases	Announce bases	Keep events with matching bases
5. Error check	Compare subset of bits	Compare subset of bits	Estimate error rate (QBER)
6. Post-process	Error correction + privacy amplification	Error correction + privacy amplification	Obtain shared secret key

5. QKD in Optical Access Networks (PON)

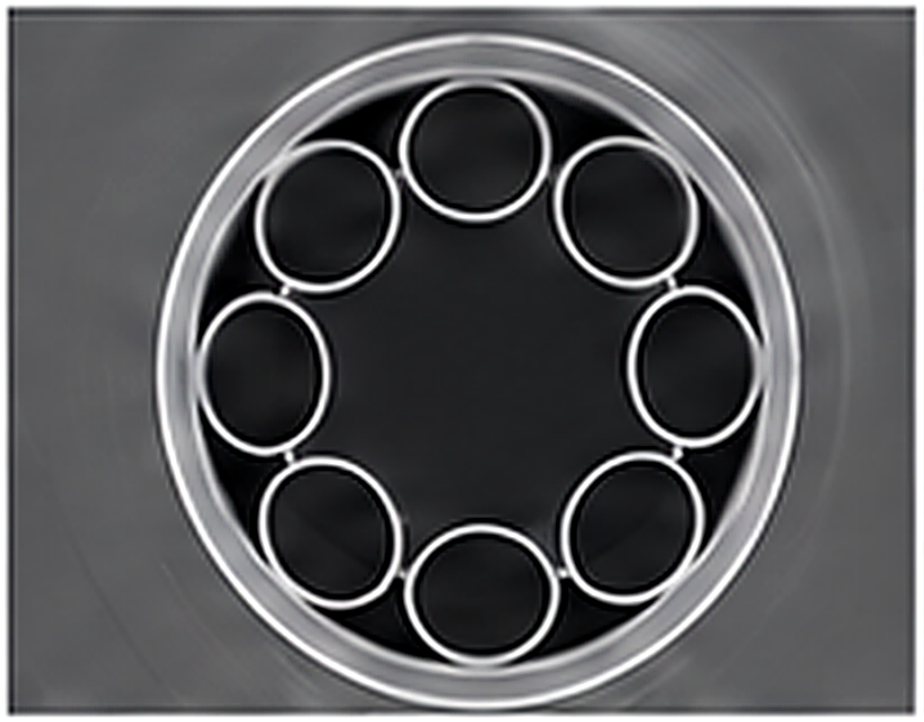
- Passive Optical Network (PON) architecture
- Shared fiber and passive splitters
- Conventional data + QKD coexistence
- Classical channels generate Spontaneous Raman Scattering (SpRS) noise
- Fiber attenuation and splitting loss affect QKD performance
- Proper wavelength allocation to reduce noise





6. Challenges and Future Directions

- Reduce Raman noise (wavelength, fiber design)
- Use low-loss and low-Raman fibers (e.g., hollow-core fibers)
- Optimize network architecture (split ratios, channel allocation)
- Increase secret-key rate and reach
- Integrate QKD with existing optical networks
- Towards practical and secure quantum communication networks



Nested antiresonant hollow-core fiber (NANF)

